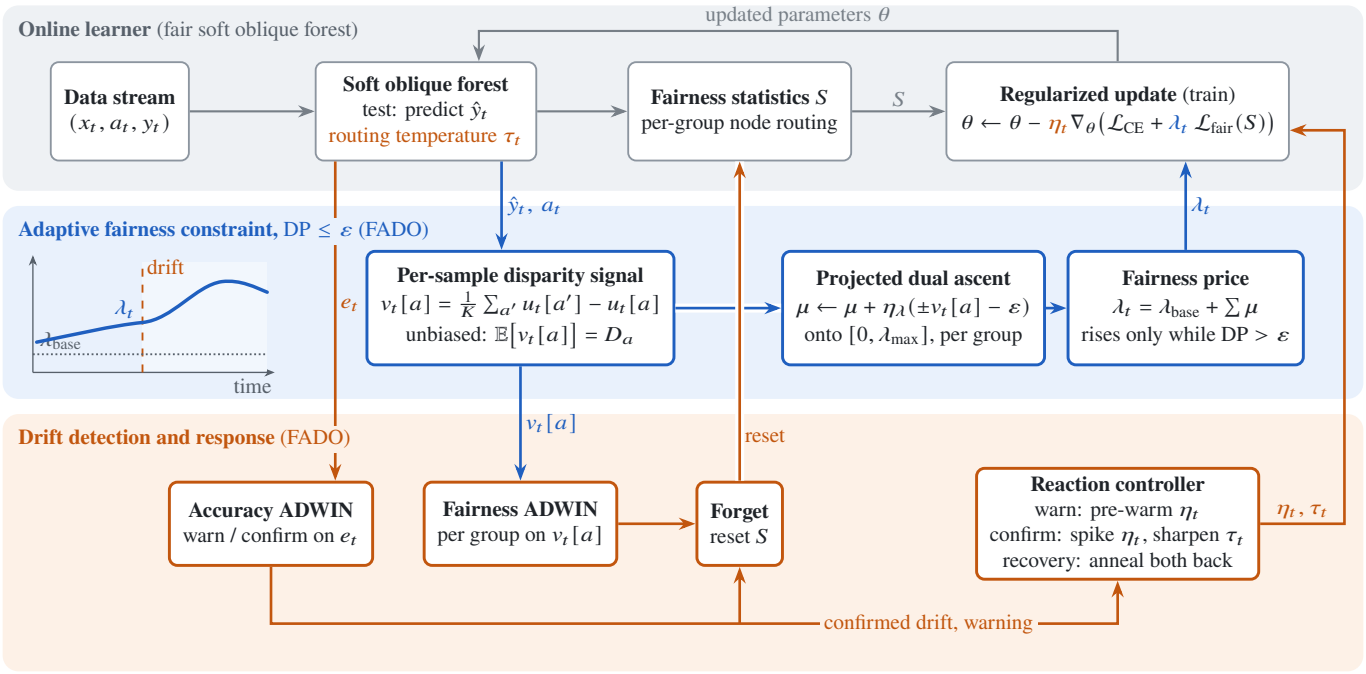

Online learner (fair soft oblique forest)
updated parameters θ
Data stream
(x_t, a_t, y_t)
Soft oblique forest
test: predict $\hat{y}_t$
routing temperature τ_t
Fairness statistics S
per-group node routing
S
Regularized update (train)
$\theta \leftarrow \theta - \eta_t \nabla_\theta (\mathcal{L}_{\text{CE}} + \lambda_t \mathcal{L}_{\text{fair}}(S))$
$\hat{y}_t,\ a_t$
λ_t
Adaptive fairness constraint, DP $\leq \varepsilon$ (FADO)
drift
λ_t
λ_{base}
time
e_t
Per-sample disparity signal
$v_t[a] = \frac{1}{K}\sum_{a'} u_t[a'] - u_t[a]$
unbiased: $\mathbb{E}[v_t[a]] = D_a$
Projected dual ascent
$\mu \leftarrow \mu + \eta_\lambda(\pm v_t[a] - \varepsilon)$
onto $[0, \lambda_{\max}]$, per group
Fairness price
$\lambda_t = \lambda_{\text{base}} + \sum \mu$
rises only while DP > ε
$v_t[a]$
reset
Drift detection and response (FADO)
Accuracy ADWIN
warn / confirm on e_t
Fairness ADWIN
per group on $v_t[a]$
Forget
reset S
Reaction controller
warn: pre-warm η_t
confirm: spike η_t, sharpen τ_t
recovery: anneal both back
η_t, τ_t
confirmed drift, warning